

## 1.2 Evolution of Computers

This chapter discusses the history and the development of the computers. First the chapter gives you an idea about the history of computers. Next it given an explanation about modern day computers. In order to have a complete understanding about computer systems, it is quite important to know about how the computer has evolved from its inception.

### 1.2.1 Key Developments

Many new ideas have contributed significantly to the development of computer systems. Some of the key developments of computer systems for the past are:

- **Abacus (3000BC):** the first counting device was the abacus, originally from Asia. It worked on a place-value notion meaning that the place of a bead or rock on the apparatus determined how much it was worth.
- **Napier's bones (1600):** Napier's bones is a manually-operated calculating device created by John Napier of Merchiston for calculation of products and quotients and numbers.
- **Pascaline (1642):** Blaise pascal invented the mechanical calculator called Pascaline. This calculating machine could add and subtract two numbers directly and multiply and divide by repetition.
- **Punch cards (1800):** In France, Joseph Marie Jacquard invents a loom that uses punched wooden cards to automatically weave fabric designs. Early computers would use similar punch cards.
- **Difference Engine (1812):** Charles P. Babbage, the "father of the computer" designed a machine, the difference engine which would be steam powered, fully automatic and commanded by a fixed instruction program.
- **Ada Lovelace (1840):** Ada Lovelace, world's first computer programmer, provided the first algorithm intended to be processed by Charles Babbage's early mechanical general-purpose computer, the Analytical Engine. She suggested that a binary system should be used for storage rather than a decimal system.
- **Boolean logic (1850):** George Boole developed Boolean logic which would later be used in the design of computer circuitry.



- **Hollerith's Tabulator (1890):** Dr. Herman Hollerith introduced the first electromechanical, punched card data-processing machine which was used to compile information for the 1890 U.S. census. Hollerith's tabulator became so successful that he started his own business to market it (IBM).
- **Vacuum Tube (1906):** The vacuum tube is invented by American physicist Lee De Forest.
- **Turing Machine (1943):** British mathematician Alan Turing developed a hypothetical device, the Turing machine which would be designed to perform logical operation and could read and write.
- **Mark 1 (1944):** Conceived by Harvard physics professor Howard Aiken, and designed and built by IBM, the Harvard Mark 1 is a room-sized, relay-based calculator. This machine could handle all four arithmetic operations, and had special built-in programs for logarithms and trigonometric functions. The machine had a fifty-foot long camshaft running the length of machine that synchronized the machine's thousands of component parts and used 3,500 relays.
- **ENIAC-Electrical Numerical Integrator and Calculator (1946):** The giant ENIAC machine was developed by John W. Mauchly and J. Presper Eckert at the University of Pennsylvania.
- **UNIVAC-Universal Automatic Computer (1951):**
- **EDVAC-Electronic Discrete Variable Automatic Computer:** Unlike its predecessor the ENIAC, it was binary rather than decimal, and was designed to be a stored-program computer. ENIAC inventors John Mauchly and J. Presper Eckert proposed the EDVAC's construction in August 1944.
- **Transistor:** William Shockley, John Bardeen and Walter Brattain of Bell Laboratories invent the transistor. They discovered how to make an electric switch with solid materials and no need for a vacuum.

- **First Generation (1951-1958)**
  - Used vacuum tubes for internal operations.
  - Used low level languages for programming (machine language)
  - Used magnetic drums for primary memory
  - Primary memory was limited.
  - Heat and maintenance problems were there
  - Used punch cards for inputs and outputs
  - Input and output were slow.

ENIAC, EDSAC, IBM 701, and Manchester Mark 1.

- **Second Generation (1958-1964)**
  - Used transistors for internal operations
  - Increased use of high level languages
  - Used magnetic cores for primary memory
  - Increased memory capacity
  - Binary coded data were used.
  - Increased processing speed



- Used magnetic tapes and disks for secondary storage  
IBM 1620, UNIVAC 1108

- **Third Generation (1965-1970)**

- Used ICs on silicon chips for internal operations
- Memory capacity was increased
- Minicomputers became a common use
- Software industry emerged
- Reduction in size and cost of computers happened
- Increase in speed and reliability

Honeywell-6000 series

- **Fourth Generation (1972-2010)**

- Used large scale integration (LSI) and very large-scale integration (VLSI) for operations
- Development of the microprocessor happened
- Introduction of micro and super computers
- Increase in speed, power and storage capacity
- Parallel processing was introduced
- Robotics was introduced
- Increased use of Micro/personal computers

Apple, IBM

- **Fifth Generation (2010 to Present)**

- Artificial intelligence and expert systems were introduced
- Symbolic manipulation and symbolic reasoning are required

IBM Watson, Microsoft Cortana, Apple Siri

| Generations                              | Major Hardware Technology                                                                                                                       | Software Used                                                                                                                       | Characteristics                                                                                                                                                                                                       | Systems invented                                                                                                                  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| First Generation Computers<br>1940-1956  | <ul style="list-style-type: none"> <li>• Vacuum Tubes</li> <li>• Punch Cards are used for input, process, output and storage of data</li> </ul> | <ul style="list-style-type: none"> <li>• Machine language</li> <li>• Assembly language</li> <li>• Stored Program Concept</li> </ul> | <ul style="list-style-type: none"> <li>• High Heat Generation</li> <li>• Slow in processing</li> <li>• Large in size</li> <li>• Not Portable</li> <li>• Consumes a lot of electricity</li> <li>• Expensive</li> </ul> | ENIAC<br>EDVAC<br>EDSAC<br>UNIVAC<br>IBM 701                                                                                      |
| Second Generation Computers<br>1956-1963 | <ul style="list-style-type: none"> <li>• Transistors</li> <li>• Tape</li> <li>• Floppy Disk, Tape for Secondary Storage</li> </ul>              | <ul style="list-style-type: none"> <li>• High-level Programming language</li> <li>• Use of Assembly Language</li> </ul>             | <ul style="list-style-type: none"> <li>• Smaller in size.</li> <li>• Less heat Generation</li> <li>• Low power consumption</li> <li>• Comparatively faster than the first generation</li> <li>• Expensive</li> </ul>  | <ul style="list-style-type: none"> <li>• Honey well 400</li> <li>• IBM 7030</li> <li>• CDC 1604</li> <li>• UNIVAC LARC</li> </ul> |

| Generations                                | Major Hardware Technology                                                                                                                                                                                                                                                                                                                            | Software Used                                                                                                                                                                           | Characteristics                                                                                                                                                                                                                 | Systems invented                                                                                                       |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Third Generation Computers<br>(1964-1975)  | <ul style="list-style-type: none"> <li>• Integrated Circuits (IC)</li> <li>• High capacity disks for secondary storage</li> <li>• Keyboard and mouse for data input</li> </ul>                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• birth of Operating Systems (OS)</li> <li>• Well developed Programming languages</li> <li>• high level computer languages for coding</li> </ul> | <ul style="list-style-type: none"> <li>• Smaller in size</li> <li>• Less heat</li> <li>• Generation</li> <li>• Comparatively faster than the second generation</li> <li>• Expensive</li> <li>• Low power consumption</li> </ul> | <ul style="list-style-type: none"> <li>• IBM-360/370</li> <li>• PDP-8</li> <li>• PDP-11</li> <li>• CDC 6600</li> </ul> |
| Fourth Generation Computers<br>(1975-1989) | <ul style="list-style-type: none"> <li>• LSIC (Large Scale Integrated Circuits) and VLSIC (Very Large Scale Integrated Circuits)</li> <li>• Microprocessor</li> <li>• Palm Tops</li> <li>• High Capacity hard disks</li> <li>• Floppy disk</li> <li>• Optical disk</li> <li>• Personal computers (PC)</li> <li>• Faster computer networks</li> </ul> | <ul style="list-style-type: none"> <li>• OS with GUI (Graphical User Interface)</li> <li>• UNIX OS</li> </ul>                                                                           | <ul style="list-style-type: none"> <li>• Very small in size</li> <li>• Portable</li> <li>• Upgradable</li> </ul>                                                                                                                | <ul style="list-style-type: none"> <li>• IBM PC</li> <li>• Apple II</li> </ul>                                         |



| Generation                                    | Major Hardware Technology                                                                                                                                               | Software Used                                                                                                                                                                                                                                                                                              | Characteristics                                                                                                                                                                                    | Systems invented                                                                                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Fifth Generation Computers (1989 to present ) | <ul style="list-style-type: none"> <li>• ULSI (Ultra Large Scale Integration)</li> <li>• Very High Capacity Hard disks and optical disks</li> <li>• Internet</li> </ul> | <ul style="list-style-type: none"> <li>• Operating Systems with GUI (Graphical user Interface)</li> <li>• Internet and multi-media applications</li> <li>• Voice recognition based on AI (Artificial Intelligence)</li> <li>• Character recognition</li> <li>• Hand-writing recognition systems</li> </ul> | <ul style="list-style-type: none"> <li>• Portable</li> <li>• Less Expensive</li> <li>• Smaller in size</li> <li>• Easy operation</li> <li>• High reliability</li> <li>• High efficiency</li> </ul> | <ul style="list-style-type: none"> <li>• IBM notebooks</li> <li>• Pentium PCs</li> <li>• SUN workstations</li> </ul> |

## Summary

- Images, words, numbers or symbols which do not convey a meaning when standing alone are called data.
- Information is obtained by processing data. Information is used to make decisions.
- Computer is a system.
- Computers and computer based systems are used to process data.
- Providing data is called 'input' while extracting information is called 'output'.
- Quality of information is important, (relevancy, completeness, accuracy, timeliness, low cost)
- Data input, processing and data output are components of a system.
- Technology is used to exchange the processed data. This is called ICT.
- Applications of ICT have made man's life easy. There are many applications such as e-Government, education, health, agriculture, business, transport, entertainment etc.
- The advent of computing dates back to 5000 years. Computing is categorized into many generations from the period of automation.